

Advay Kadam

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EDUCATION

University of Illinois Urbana-Champaign

Champaign, IL

B.S. Computer Science and Statistics, Minor in Mathematics | GPA: 3.87

Expected Graduation: December 2026

- **Relevant Courses:** Algorithms and Models of Computation, Machine Learning, Abstract Linear Algebra, Time Series Analysis, Statistical Modeling, Statistics and Probability, Data Structures and Algorithms

TECHNICAL SKILLS

Languages: Python, C++, SQL, R, Java, JavaScript, HTML/CSS

Tools: PyTorch, Tensorflow, NumPy, Pandas, Flask, Azure, AWS SageMaker, Apache Spark/PySpark, Matplotlib, Git

EXPERIENCE

Bank of New York Mellon

June 2025 – Present

Data Science and Engineering Intern

Jersey City, NJ

- Developed pipeline integrating logistic regression with a RAG agent to automate metadata security classification across **4,000+** datasets, improving tagging accuracy by **10%** and reducing manual audit time by **15 hours/week**
- Performed NLP-driven feature engineering with Pandas/NumPy; applied TF-IDF and sentence embeddings using scikit-learn to vectorize unstructured text fields for classification

Procter & Gamble

May 2024 – Aug 2024

Software Engineering Intern

Cincinnati, OH

- Optimized SQL transformations and added DAG-aware caching in PySpark, cutting sales ETL pipeline refresh time by **60%** on **200k+** records; accelerated dashboard updates used by analysts to drive **\$300M+** in revenue
- Implemented statistical anomaly detection using Isolation Forests with Python to flag deviations in batch loads
- Fine-tuned YOLOv9 model on retail image data to **95%** accuracy and executed model on Azure ML pipeline

RESEARCH AND CAMPUS INVOLVEMENT

Siebel Center of Computing and Data Science

Jan 2025 – Present

Course Assistant for Data Structures and Algorithms

Urbana, IL

- Led office hours and discussion sessions on abstract data types, C++ algorithms, and complexity analysis
- Developed and maintained algorithmic exercises (e.g., nearest neighbor search) for **400+** enrolled students

Disruption Lab at Gies

Aug 2024 – May 2025

Software Engineer

Urbana, IL

- Led a team of 6 students to train transfer learning model to detect AI-generated deepfake images with TensorFlow
- Trained and deployed model in AWS SageMaker, boosting accuracy to 96% via automated hyperparameter tuning

National Center for Supercomputing Applications

Aug 2023 – May 2024

Undergraduate Researcher

Urbana, IL

- Analyzed 2PB+ of real-time network metadata using Pandas, NumPy, Shell scripting to assess PQC resilience
- Identified that over **99%** of SSH traffic lacked quantum-resistant encryption in collaboration with Dr. Phuong Cao
- Developed Flask dashboard to parse Zeek logs and deliver live metrics on PQC algorithm usage and vulnerability

Rice University

Nov 2021 – Mar 2023

Research Intern

Houston, TX

- Worked with Professor Aditya Mohite to enhance 2D-perovskite property predictions for photovoltaic applications
- Built PyTorch based regression model ($R^2 = 0.95$) to predict the energy-gap of 2D-halide perovskites
- Web-scraped and engineered **30+** chemical features using Pandas, NumPy, and Selenium for model training

PROJECTS

IMC Prosperity | Python, Pandas, Matplotlib, Seaborn

- Placed in the top **3%** in a 15-day global trading challenge, writing Python scripts for algorithmic trading
- Applied Backtesting and implemented strategies related to market making, Black-scholes pricing, and regression

Regression Library | C++, Xtensor, Boost, CMake

- Built C++ library for multiple linear regression and hypothesis testing for reduced regression coefficients
- Implemented stochastic/batch gradient descent algorithms and confidence intervals for true parameter estimation

Time Series Regression Analysis of Sea Surface Temperatures | R

- Performed research on creating robust regression models for forecasting sea surface temperatures during El Niño
- Modeled autocorrelated errors and validated residual assumptions via ACF/PACF plots and Ljung-Box tests

Datathons | 3rd place at UIUC Spring 2024 Datathon, 2nd place at UIUC Fall 2023 Datathon